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Larger Achilles and plantar fascia induce lower duty factor during barefoot running

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ABSTRACT

Objectives: Tendons play a crucial role allowing the storage and release of mechanical energy during the running cycle. Running kinematics, including duty factor, constitute a basic element of the runner's biomechanics, and can determine their performance. This study aimed to analyze the link between Achilles tendon and plantar fascia morphology and running parameters, considering the influence of wearing shoes versus running barefoot.
Design: Cross-sectional study.
Methods: 44 participants (30 men and 14 women) engaged in two running sessions, one with shoes and one without, both lasting 3 min at a consistent speed of 12 km/h. We captured running kinematic data using a photoelectric cell system throughout the sessions. Before the trials, we measured the thickness and cross-sectional area of both the Achilles tendon and plantar fascia using ultrasound.
Results: The Pearson test revealed a significant correlation ($p < 0{,}05$) between Achilles tendon and plantar fascia morphology and contact time ($r > -0.325$), flight time ($r > -0.325$) and duty factor (ratio of ground contact to stride time) ($r > -0.328$) during barefoot running. During the shod condition, no significant correlation was found between connective tissue morphology and kinematic variables.
Conclusions: In barefoot running, greater size of the Achilles tendon and plantar fascia results in a reduced duty factor, attributed to longer flight times and shorter contact times.
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Practical implications

- Larger AT and PF result in shorter CT, longer FT, and lower DF during barefoot running.
- Aerial runners might enhance performance by training to strengthen the Achilles tendon and plantar fascia for efficiency.
- Biomechanics should be a primary consideration when selecting a runner's training regimen.

1. Introduction

The lower limb behaves like a spring that compresses and decompresses during running.[1] This spring-like quality allows the storage of mechanical energy during the eccentric phase of the stance and its subsequent as elastic energy during the concentric phase (propulsion), facilitating subsequent movements.[2] The stretch–shortening cycle is a key neuromuscular mechanism for this storage-release energy process.[3] While muscles might appear to be the primary actors in the stretch–shortening cycle, it has been suggested that without proper tendon function, muscle shortening speed would need to increase during rebounding actions like jumping or running. Consequently, this would result in a less efficient energy storage and release mechanism, leading to reduced performance..[4]

Tendons consist primarily of collagen (comprising 65–80 %) and elastin, which are embedded within a proteoglycan–water matrix.[5] In addition to its significant role in the stretch–shortening cycle, recent findings from a systematic review underscore the relevance of lower limb tendon morphology in running biomechanics.[6] Indeed, runners tend to have larger tendons compared to untrained individuals,[6] and a positive correlation has been established between the Achilles tendon cross-sectional area (AT-CSA) and vertical stiffness.[7] While numerous studies have examined tendon morphology and running spatiotemporal variables (i.e., flight time [FT], ground contact time [CT], stride length [SL], step frequency [SF]),[7,8] none have investigated the potential correlation between these factors. Furthermore, no study has incorporated

Abbreviations: AT, Achilles tendon; PF, plantar fascia; CSA, cross-sectional area; CT, contact time; FT, flight time; SF, step frequency; SL, stride length; DF, duty factor.

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duty factor (DF) in its kinematic analyses. Duty factor, defined as the ratio of CT and the total stride time (i.e., CT + FT), offers a more comprehensive understanding of the overall running pattern compared to examining these temporal elements in isolation.[9] It serves as an objective parameter reflecting the time spent in ground contact during running.[10] Although it has been considered valuable for assessing energy expenditure during races, research has yielded conflicting results regarding its impact on running economy.[11]

The most frequently assessed morphological characteristics of connective tissue are its cross-sectional area (CSA) and thickness.[12] Ultrasound and nuclear magnetic resonance imaging techniques are commonly used for their proven reliability.[12] In the realm of studying tendon morphology in runners, the majority of research has concentrated on the Achilles tendon (AT).[6] This focus may stem from the established significance of the AT in influencing running economy..[13] Fletcher et al. proposed that the stiffness of the AT is a central mechanism underlying improved running economy.[13] Qiao and Jindrich[14] characterized joint function during running into four distinct behaviors: (1) a motor that generates positive work; (2) a spring that stores and releases elastic deformation energy; (3) a strut that generates large force with small changes in length; and (4) a shock absorber that elongates to absorb energy. In these four phases the AT has a fundamental role, especially in phases 2 and 4.[15] While the plantar fascia (PF) is not typically classified as a tendon, certain studies view it as an extension of the AT due to their shared attachment point at the calcaneus bone, where it assists in the AT's function.[16] Consequently, when assessing running biomechanics, the behavior of both AT and PF must be taken into consideration.[6]

It has previously been observed how running with or without shoes influences running biomechanics.[17] A review of biomechanical differences between shod running and barefoot running found moderate evidence that barefoot running is associated with reduced peak ground reaction force, increased foot and ankle plantarflexion and increased knee flexion at ground contact compared with shod running.[18] Moreover, it seems that runners who typically land on their heels shift to a forefoot strike pattern when transitioning from running with shoes.[19] This change in foot strike pattern influences which muscles are predominantly engaged during the running motion. A rearfoot strike occurs when the foot's initial contact with the ground is at the heel or posterior section, while a forefoot strike involves the anterior region of the foot making initial contact with the ground.[20] When running with shoes, a rearfoot strike pattern places greater demand on the knee extensors.[21] In contrast, when running barefoot, a forefoot strike pattern specifically requires the ankle plantar flexors to work harder.[22]

While previous research has explored the connection between tendon morphology and running biomechanics, the relationship between AT and PF morphology and running kinematics among runners remains uncertain. This study aims to investigate the link between AT and PF morphology (including CSA and thickness) and running spatiotemporal parameters. Furthermore, we seek to determine whether the barefoot and shod running conditions impact this potential correlation. We hypothesize that, particularly during barefoot running, runners with larger tendons would exhibit increased flight time and reduced contact time due to their presumably improved capacity for storing and releasing greater elastic energy.

2. Methods

30 male and 14 female amateur endurance runners (Table 1), voluntarily participated in this study. All participants met the inclusion criteria: [i] age over 18, [ii] three or more training sessions per week,[23] and [iii] being free of any lower limb injuries in the six months preceding data collection that might affect their running biomechanics.[24] Following a comprehensive briefing on the study's objectives and procedures, every participant provided written informed consent to participate, in accordance with the ethical guidelines outlined in the World Medical Association's Declaration of Helsinki (2013). Participants were explicitly informed of their voluntary participation and the option to withdraw from the study at any point. The Institutional Review Board approved the study (University San Jorge, Zaragoza, Spain 006-18/19).

Table 1
Descriptive characteristic of the participants (mean, ±SD; n, %).

	Men (n = 30)	Women (n = 14)
Age (years)	29.1 (7.2)	26.1 (6.1)
Height (m)	1.75 (6.2)	164.8 (4.6)
Body mass (kg)	72.0 (7.3)	57.2 (6.5)
BMI ($kg*m^{-2}$)	23.3 (1.8)	21.0 (2.1)
Body fat %	13.8 (2.8)	17.2 (3.3)
Lean mass %	83.5 (3.5)	73.7 (4.1)

Based on previous studies[25,26] G-Power software Version 3.1.9.7 (University of Dusseldorf, Dusseldorf, Germany) was used to calculate the achievable power with the following specifications: exact tests through a two tail bivariate normal correlation model with correlation $\rho = 0.50$ for alternative hypothesis (H1) and $= 0$ for null hypothesis (H0), α err prob $= 0.05$. The selected sample size of 44 participants was found to provide a 94 % chance of successfully rejecting the null hypothesis of no correlation between the variables studied.

The study took place in a single session, with each participant following the same standardized protocol under researcher supervision. Before data collection, participants engaged in a structured dynamic warm-up lasting 5 min, which included exercises such as squats, lunges, and hinging movements.[27] All running conditions were completed on a motorized treadmill with a slope of 0 % (HP cosmos Pulsar 4P; HP cosmos Sports & Medical, Gmbh, Nußdorf, Germany) and kinematic data were recorded for analysis. Following the warm-up, an 8-minute adaptation program was implemented, involving an incremental increase in speed by 1 km/h each minute, reaching a final speed of 12 km/h.[28] This adaptation time was performed for each running condition, i.e., shod–barefoot, and therefore was carried out in the condition in which the running protocol was subsequently completed. Subsequently, participants ran for 3 min under the first footwear condition (either shod or barefoot) at a pace of 12 km/h.[29] To ensure representative data in healthy adults (95 % confidence intervals), a minimum of −6 and −8 strides were analyzed.[25] Following this, participants ran under the alternate footwear condition at the same speed for an additional 3 min. The sequence of shod and barefoot conditions was randomized.

High-definition ultrasound images were acquired in B-mode using a linear probe 5–16 MHz (LOGIQ S7 EXPERT, General Electric, Germany, 2013). Longitudinal and transversal views of AT and PF were captured before the running protocol. A recent review has indicated that ultrasound measurements of tendon dimensions are reliable in terms of both relative reliability and absolute reliability.[30]

To assess AT, participants assumed a prone position with both knees extended and their feet positioned outside of the bed, keeping the ankle in neutral position.[12,17,31] A reference point located 3 cm proximal to the insertion of the tendon into the calcaneus bone, measured using the ultrasound device, was used to determine both tendon thickness and CSA[12,17,31] (Fig. 1).

The assessment of the PF via ultrasound was conducted with participants in prone position, with both knees in extension, ankles in neutral position, and their toes extended against the surface of the bed.[12,17,31] To measure the thickness of the plantar fascia (PF), a reference point was identified by the ultrasound device. This point extended vertically from the anterior edge of the plantar surface of the calcaneus bone to the anterior edge of the PF (Fig. 1).[12,17,31]

For all anatomical structures, a frequency of 12 MHz and a gain of 100 dB were consistently applied. Each measurement was performed twice by a highly experienced researcher with over a decade of expertise in diagnostic ultrasound imaging. The clearest image, as determined by the examiner, was selected for subsequent calculation of morphological variables. To measure thickness and CSA, the software ImageJ (NIH,

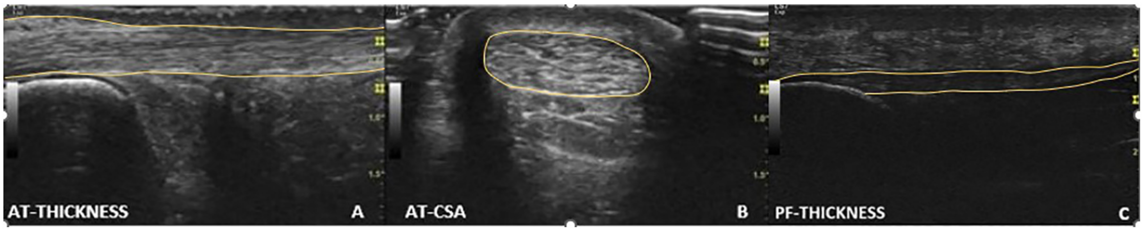

Fig. 1. Ultrasound images of AT (i.e., thickness [A] and CSA [B]) and PF [C] morphology. Thickness and CSA are highlighted in yellow. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Baltimore, MD) was utilized.[8] Here, the polygon tool was employed for CSA calculations.

In consideration of the previously established correlation between body mass and tendon morphology characteristics,[32] CSA and thickness values were normalized to one-third of the participant's body mass for statistical analysis.[33]

The spatiotemporal variables of CT, FT, SL, and SF were measured using a photoelectric cell system (OptoGait, Microgate, Bolzano, Italy), which was previously validated for the assessment of running gait spatiotemporal parameters.[34] The 2 bars of the photoelectric cell system were fixed and stabilized at both sides of the treadmill (Fig. 2). Both CT and FT values are presented in absolute terms, as well as relative to the entire running cycle (CT% and FT%, respectively). The data of the right limb were extracted at a sampling frequency of 1000 Hz, encrypted and stored on a computer. Limb dominance was not considered.[35]

Additionally, DF was calculated according to the following equation[36]:

$$DF\ (\%) = \frac{\mathrm{CT}}{(CT + FT)} * 100 \tag{1}$$

To determine the foot strike pattern, a high-definition camera (Imaging Source DFK 33UX174, The Imaging Source Europe GmbH;

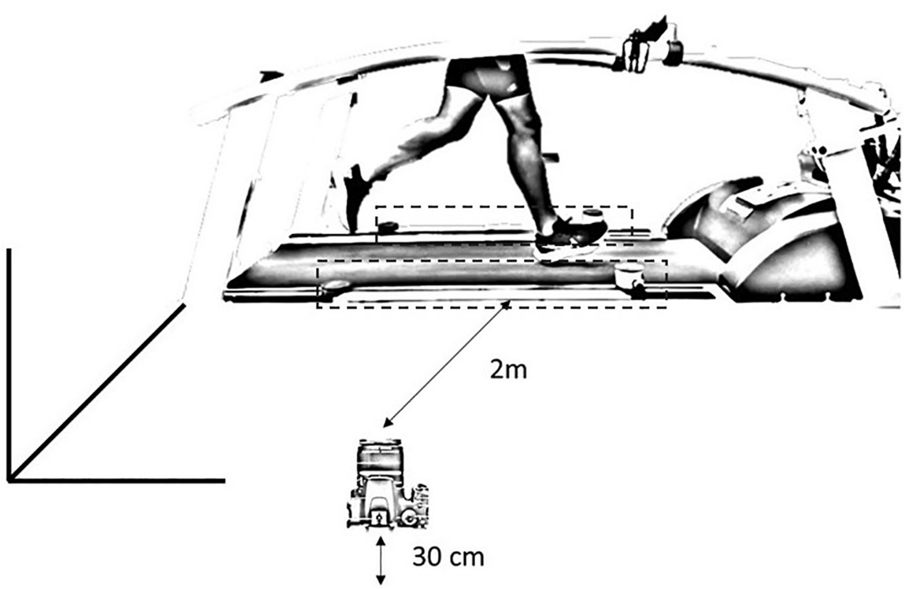

Fig. 2. Location of the photoelectric cell system, marked with dashed lines, and the position of the video camera in relation to the treadmill.

Fig. 3. A) Forefoot strike pattern during barefoot running. B) Rearfoot strike pattern during shod running.

Table 2
Relationship by reporting Pearson coefficient (95 % confidence intervals) between AT and PF morphology and running kinematics.

	Shod running						
	CT	FT	CT%	FT%	SF	SL	DF
AT-thickness	0.023 (−0.079–0.318)	0.224 (−0.079–0.488)	−0.228 (−0.491–0.075)	0.228 (−0.075–0.491)	−0.130 (−0.411–0.173)	−0.203 (−0.472–0.099)	0.207 (−0.474–0.097)
AT-CSA	0.220 (−0.082–0.486)	0.093 (−0.210–0.379)	−0.064 (−0.355–0.237)	0.064 (−0.237–0.355)	−0.251 (−0.510–0.050)	−0.082 (−0.370–0.220)	0.046 (−0.338–0.254)
PF-thickness	0.116 (−0.187–0.400)	0.153 (−0,151–0.430)	−0.141 (−0.421–0.162)	0.141 (−0.162–0.421)	−0.132 (−0.413–0.171)	-0.202 (−0.470–0.101)	−0.116 (−0.399–0.188)

AT: Achilles tendon; PF: plantar fascia; CT: contact time; FT: flight time; CT%: contact time relative to gait cycle; FT%: flight time relative to gait cycle; SF: step frequency; SL: stride length; DF: duty factor.
* $p < 0.05$.
** $p < 0.01$.

Germany) was placed 2 m lateral to the treadmill and leveled with the running surface. Videos were sampled at 240 frames/s. The foot strike pattern was determined by an experienced researcher in running biomechanics analysis through slow-motion video playback. The researcher paused the video at the moment when the foot initially made contact with the ground. Depending on whether this contact occurred at the rear part or the front part of the foot, the participant was classified as a rearfoot or forefoot striker, respectively (Fig. 3). This method has previously demonstrated both validity and reliability.[37]

Descriptive data are presented as mean and standard deviation (SD). The normality distribution of the data was confirmed by Shapiro–Wilk's test ($p > 0.05$). To determine the intra-rater reliability of the measures related to the morphology of the connective tissue, intra class correlation coefficients (ICCs) were calculated for each parameter. Additionally, the 95 % confidence interval (CI) of the ICC value was provided.[38] To analyze the relationship between the morphology of AT and PF and running kinematics, a Pearson correlation analysis was conducted for the whole group. The following criteria were adopted to interpret the magnitude of correlations between measurement variables: <0.1 (trivial), 0.1–0.3 (small), 0.3–0.5 (moderate), 0.5–0.7 (large), 0.7–0.9 (very large) and 0.9–1.0 (almost perfect).[39] All statistical analyses were performed using SPSS software version 28.0 (SPSS Inc., Chicago, IL, USA) and statistical significance was accepted at an alpha level of 0.05.

3. Results

An excellent intra-rater reliability was reported for the measures related to the morphology of the connective tissue (ICC > 0.989, 95 % CI: 0.913–0.996).

Table 2 shows the Pearson correlation analysis regarding the AT and PF morphology. Under barefoot running, a significant relationship ($p < 0.05$) between AT thickness and CT ($r = -0{,}450$), FT ($r = 0{,}431$), CT % ($r = -0{,}463$), FT % ($r = 0{,}463$), SL ($r = -0{,}335$), and DF ($r = -0{,}469$) was found. AT-CSA shows a significant relationship ($p < 0.05$) with FT ($r = 0.315$), CT % ($r = -0.325$), FT % ($r = 0.325$) and DF ($r = -0{,}328$) when barefoot running. During shod running, no significant correlation was found between AT morphology and kinematic variables ($r < 0{,}251$).

Regarding the PF morphology, during barefoot running, a significant relationship ($p < 0.05$) between PF thickness and CT ($r = -0{,}423$), FT ($r = 0{,}425$), CT % ($r = -0{,}447$), FT % ($r = 0{,}447$), SL ($r = -0{,}321$), and DF ($r = -0{,}449$) was found. When running with shoes, no significant correlation was found between PF morphology and kinematic variables ($r < 0{,}202$).

Regarding the foot strike pattern, 71 % (n = 32) of the runners showed a rearfoot strike pattern and 29 % (n = 12) showed a forefoot strike pattern under the shod condition. When barefoot, 69 % (n = 30) of the participants showed a forefoot strike pattern and 31 % (n = 14) showed a rearfoot strike pattern.

4. Discussion

The purpose of the present study was to investigate the relation between the AT and PF morphology (i.e., thickness and CSA) and running kinematics. Additionally, it was sought to determine whether the shod or barefoot condition influences this potential relationship.

This study mainly found a significant correlation between the morphological characteristics of AT and PF and CT and FT, for both absolute and relative values, during barefoot running. Additionally, a significant correlation was also found between AT and PF morphology and DF while barefoot running. The main findings of this study support our initial hypothesis. Specifically, during barefoot running, individuals with larger tendons demonstrated a higher flight time and a lower contact time, likely due to their improved capacity to store and release greater elastic energy. The shod–barefoot condition was revealed to be a crucial factor in the relationship between tendon morphology and running kinematics. However, this relationship was only evident during barefoot running, with no significant correlation observed during shod running. This finding aligns with previous research[17] and can be attributed to the fact that runners tend to adopt a rearfoot strike pattern during shod running, which places less demand on the Achilles tendon.

The biomechanical aspects of barefoot running shed light on its impact on the lower limb. This running style, characterized by a forefoot strike pattern, places unique demands on the AT. In barefoot running, there's a greater need for impulse and loading rates on the AT.[22] These specific demands on the AT may clarify the association between larger tendons and the observed higher FT and lower CT. Larger tendons appear to possess an enhanced capacity for storing and releasing elastic energy during the stretch–shortening cycle, aligning with the requirements of barefoot running.

Although no study has directly explored the link between tendon morphology and running kinematics, previous research on lower-limb stiffness (i.e., leg stiffness and vertical stiffness) has shown similar results.[7] Noteworthy, lower limb stiffness serves as a valuable indicator for assessing the storage–recoil mechanism of elastic energy during running. As demonstrated by Monte et al. in half-marathon runners, a higher AT-CSA was correlated with greater vertical stiffness ($r = 0{,}45$) and leg stiffness ($r = 0{,}36$) in faster runners.[7] However, to the best of the authors' knowledge, the aforementioned study did not investigate the foot strike pattern adopted by these runners. The findings of this study align with a prior investigation, offering additional support to the reported results.[7,17] One of the previous studies[17] examined the relationship between the morphology of major lower-limb tendons (i.e., AT, PF and patellar tendon) and lower limb stiffness in long-distance runners. Their study considered both shod and barefoot conditions and took foot strike patterns into account.[17] Notably, the results revealed the influence of running with or without shoes, with its consequent biomechanical differences. Under shod conditions, there was a trend toward a positive correlation between stiffness and patellar tendon morphology ($r > 0.124$), but it was not statistically significant. However, during barefoot running, this correlation changed, becoming

Barefoot running						
CT	FT	CT%	FT%	SF	SL	DF
−0.450** (−0.658 to −0.175)	0.431** (0.154 to −0.646)	−0.463** (−0.668 to −0.192)	0.463** (0.192–0.668)	0.042 (−0.258–0.334)	−0.335* (−0.575 to −0.042)	−0.469** (−0.672 to −0.199)
−0.274 (−0.527–0.026)	0.315* (0.022–0.561)	−0.325* (−0.567 to −0.031)	0.325* (0.031–0.567)	−0.046 (−0.338–0.255)	−0.231 (−0.494–0.071)	−0.328* (−0.570–0.035)
−0.423** (−0.638 to −0.142)	0.425** (0.148–0.642)	−0.447** (−0.657 to −0.174)	0.447** (0.174–0.657)	0.022 (−0.277–0.317)	−0.321* (−0.564 to −0.026)	-0.449** (−0.658 to −0.176)

significant with respect to the Achilles tendon and plantar fascia morphology ($r > 0.041$). It's worth mentioning that these correlations were not statistically significant, possibly due to the small sample size ($n = 14$). In contrast, the present study demonstrated statistically significant correlations for all the mentioned relationships, providing greater consistency in the findings. The larger sample size ($n = 44$) in this study may explain the differences between the two studies.

Based on these findings, it is advisable for future studies to take into account the foot strike pattern exhibited by runners and how it might alter their running biomechanics when investigating the impact of tendon morphology. Regarding DF, our study revealed a significant correlation between AT and PF morphology and DF during barefoot running. Our results confirm that larger AT and PF dimensions correlate with a lower DF ($r > -0.328$). Some previous studies have categorized runners based on their DF, distinguishing between "terrestrial" runners with higher DF and "aerial" runners with lower DF characteristics.[11,40] Aerial runners tend to favor vertical oscillations and longer flight times (FTs), while terrestrial runners prioritize propelling their bodies forward with shorter FT. While a lower CT and a higher FT have historically been associated with reduced energy expenditure during running,[41] there is currently some debate on this matter.[11] It seems that runners tend to adopt running patterns that optimize their energy efficiency individually.[42] Then, to date, no clear association has been established between energy cost and DF.[11] Furthermore, DF has been shown to be to be speed-dependent, with an inverse relationship as running speed increases. As running velocity rises, DF tends to decrease.[11] Lussiana et al. stated that the decrease in DF at higher speeds may be attributed to reduced muscle stretching and shortening during the contact phase, while tendon stretching and shortening increase.[11] This occurs because muscle force intensifies with speed, leading to greater elastic energy storage and recoil.[11] This biomechanical process aligns with the results of our study, which indicate that DF decreases, especially when the AT and PF are subjected to higher demands, as seen during barefoot running.

Aerial runners optimize their energy expenditure by increasing FT and vertical oscillation of the body's center of mass, which often leads to a forefoot strike pattern. In light of the findings from this study and considering that forefoot strikers place greater demands on the AT and PF, it would be valuable to tailor a runner's training regimen to align with their unique running biomechanics. This personalized approach can help optimize the functioning of their connective tissue and potentially enhance their performance.

In the last decade, some coaches have advocated for a shift toward a more forefoot strike pattern as opposed to a rearfoot strike pattern. However, it's essential to recognize that promoting a particular strike pattern can be problematic if it's not tailored to the individual runner's biomechanics. As mentioned earlier, runners tend to naturally adopt the running pattern that minimizes their energy expenditure.[11,42] Therefore, it is the training regimen that should adapt to the runner's biomechanics, not the other way around. In light of the influence of Achilles tendon (AT) and plantar fascia (PF) morphology identified in this study, aerial runners may benefit from exercises aimed at enhancing the thickness and morphology of their AT and PF. These improvements can favor their running kinematics, and plyometric training is one approach that can be considered for achieving this.[43]

Despite the findings reported here, there are some limitations to be considered. Participants wore their own running shoes, which enhanced the ecology of the study. The use of the same shoe model could limit the possible bias derived from the characteristics of the shoe, but it might jeopardize their usual performance. Additionally, the study protocol involved running at a constant speed of 12 km/h. Future research could explore how variations in running speed may influence the relationship between tendon morphology and running kinematics.

5. Conclusion

This study demonstrates that in the context of barefoot running, individuals with larger AT and PF tend to exhibit shorter CT, longer FT, and consequently, a lower DF. Although the aforementioned correlation is only found when running barefoot and this is not the condition athletes train and perform, it would be interesting for aerial runners that often show a forefoot strike pattern and optimize their running economy by extending FT, to consider incorporating training strategies aimed at adapting and strengthening their connective structures, such as AT and PF. This approach could potentially enhance their running biomechanics and overall running performance.

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Confirmation of ethical compliance

The study followed the World Medical Association's Declaration of Helsinki (2013). The study was approved by the Institutional Review Board (University San Jorge, Zaragoza, Spain 006-18/19).

CRediT authorship contribution statement

Alberto Rubio-Peirotén: Conceptualization, Methodology, Data curation, Formal analysis, Writing - Original draft preparation, Writing - Review & editing. **Antonio Cartón-Llorente:** Data curation, Writing - Review & editing. **Luis Enrique Roche-Seruendo:** Methodology, Writing - Reviewing and editing, Supervision. **Diego Jaén-Carrillo:** Methodology, Data curation, Writing - Review & editing, Supervision.

Declaration of interest statement

The present study does not have any conflict of interest.

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